

Ch25: 2, 7, 17, 21, 39, 49, 52

Basic score 4` + problem 17 2`+ problem 49 2`+ problem 52 2`

2. (a) The potential difference across C_1 (the same as across C_2) is given by

$$V_1 = V_2 = \frac{C_3 V}{C_1 + C_2 + C_3} = \frac{(15.0 \mu\text{F})(75\text{V})}{10.0 \mu\text{F} + 5.00 \mu\text{F} + 15.0 \mu\text{F}} = 37.5\text{V}.$$

Also, $V_3 = V - V_1 = V - V_2 = 75\text{ V} - 37.5\text{ V} = 37.5\text{ V}$. Thus,

$$q_1 = C_1 V_1 = (10.0 \mu\text{F})(37.5\text{ V}) = 3.75 \times 10^{-4}\text{ C}$$

$$q_2 = C_2 V_2 = (5.0 \mu\text{F})(37.5\text{ V}) = 1.88 \times 10^{-4}\text{ C}$$

$$q_3 = q_1 + q_2 = 5.63 \times 10^{-4}\text{ C}$$

(b) The potential difference V_3 was found in the course of solving for the charges in part (a). Its value is $V_3 = 37.5\text{ V}$.

(c) The energy stored in C_3 is $U_3 = C_3 V_3^2 / 2 = 1.05 \times 10^{-2}\text{ J}$

(d) From part (a), we have $q_1 = 3.75 \times 10^{-4}\text{ C}$, and

(e) $V_1 = 37.5\text{V}$ as shown in (a).

(f) The energy stored in C_1 is

$$U_1 = C_1 V_1^2 / 2 = 7.03 \times 10^{-3}\text{ J}$$

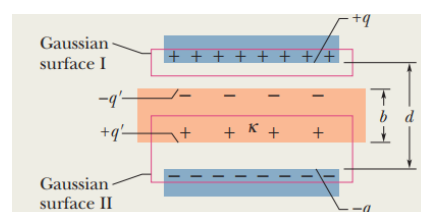
(g) Again, from part (a), $q_2 = 1.85 \times 10^{-4}\text{ C}$.

(h) $V_2 = 37.5\text{V}$ as shown in (a).

(i) The energy stored in C_2 is $U_2 = C_2 V_2^2 / 2 = 3.52 \times 10^{-3}\text{ J}$

7. (a) Initially, the capacitance is

$$C_0 = \frac{\epsilon_0 A}{d} = \frac{(8.85 \times 10^{-12}\text{ C}^2/\text{N} \cdot \text{m}^2)(0.08\text{ m}^2)}{1.2 \times 10^{-2}\text{ m}} = 59\text{ pF}.$$



(b) Assume the charge on the positive plate is q and then the electric field in the region between the vacuum and the dielectric and the electric field inside the dielectric can be get from Gauss' law.

For the Gaussian surface I and II, we have

$$\text{Gaussian surface I: } EA = \frac{q}{\epsilon_0} \Rightarrow E_1 = \frac{q}{A\epsilon_0}$$

$$\text{Gaussian surface II: } \kappa EA = \frac{q}{\epsilon_0} \Rightarrow E_2 = \frac{q}{\kappa A\epsilon_0}$$

So the potential difference between the two plates is:

$$V = E_1(d - b) + E_2b$$

And the capacitance of the capacitor is given by the definition:

$$\begin{aligned} C = \frac{q}{V} &= \frac{q}{E_1(d - b) + E_2b} = \frac{q}{\frac{q}{A\epsilon_0}(d - b) + \frac{q}{\kappa A\epsilon_0}b} = \frac{\kappa A\epsilon_0}{\kappa(d - b) + b} \\ &= \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.08 \text{ m}^2)(4.8)}{(4.8)(1.2 - 0.40) \times 10^{-2} \text{ m} + 0.4 \times 10^{-2} \text{ m}} = 80 \text{ pF} \end{aligned}$$

(c) Before the insertion, $q = C_0V = (59 \text{ pF})(120 \text{ V}) = 7.08 \text{ nC}$.

(d) Since the battery is disconnected, q will remain the same after the insertion of the slab, with $q = 7.08 \text{ nC}$.

(e) $E = q / \epsilon_0 A = 7.08 \times 10^{-9} \text{ C} / (8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(0.08 \text{ m}^2) = 10 \text{ kV/m}$.

(f) $E' = E/\kappa = (10 \text{ kV/m})/4.8 = 2.08 \text{ kV/m}$.

(g) The potential difference across the plates is

$$V = E(d - b) + E'b = (10 \text{ kV/m})(1.2 \text{ cm} - 0.4 \text{ cm}) + (2.08 \text{ kV/m})(0.4 \text{ cm}) = 88.3 \text{ V}.$$

(h) The work done is

$$W_{\text{ext}} = \Delta U = \frac{q^2}{2} \left(\frac{1}{C} - \frac{1}{C_0} \right) = \frac{(7.08 \times 10^{-9} \text{ C})^2}{2} \left(\frac{1}{80 \times 10^{-12} \text{ F}} - \frac{1}{59 \times 10^{-12} \text{ F}} \right) = -1.12 \times 10^{-7} \text{ J}.$$

17. (a) Let q be the charge on the positive plate. Since the capacitance of a parallel-plate capacitor is given by $\epsilon_0 A/d_i$, the charge is $q = CV = \epsilon_0 AV_i/d_i$. After the plates are pulled apart, their separation is d_f and the potential difference is V_f . Then $q = \epsilon_0 AV_f/d_f$ and

$$V_f = \frac{d_f}{\epsilon_0 A} q = \frac{d_f}{\epsilon_0 A} \frac{\epsilon_0 A}{d_i} V_i = \frac{d_f}{d_i} V_i.$$

With $d_i = 8.00 \times 10^{-3} \text{ m}$, $V_i = 16.0 \text{ V}$, and $d_f = 3.00 \times 10^{-3} \text{ m}$, we have $V_f = 6.0 \text{ V}$.

0.5 分

(b) The initial energy stored in the capacitor is

$$U_i = \frac{1}{2} C V_i^2 = \frac{\epsilon_0 A V_i^2}{2 d_i} = \frac{(8.85 \times 10^{-12} \text{ C}^2/\text{N} \cdot \text{m}^2)(8.50 \times 10^{-4} \text{ m}^2)(16.00 \text{ V})^2}{2(8.00 \times 10^{-3} \text{ m})} = 1.20 \times 10^{-10} \text{ J}.$$

0.5 分

(c) The final energy stored is

$$U_f = \frac{1}{2} \frac{\epsilon_0 A}{d_f} V_f^2 = \frac{1}{2} \frac{\epsilon_0 A}{d_f} \left(\frac{d_f}{d_i} V_i \right)^2 = \frac{d_f}{d_i} \left(\frac{\epsilon_0 A V_i^2}{2 d_i} \right) = \frac{d_f}{d_i} U_i. \quad 0.5 \text{ 分}$$

With $d_f / d_i = 3.00 / 8.00$, we have $U_f = 4.51 \times 10^{-11} \text{ J}$.

(d) The work done to pull the plates apart is the difference in the energy:

$$W = U_f - U_i = -7.49 \times 10^{-11} \text{ J}. \quad 0.5 \text{ 分}$$

21. The capacitance of a cylindrical capacitor is given by

$$C = \kappa C_0 = \frac{2\pi\kappa\epsilon_0 L}{\ln(b/a)},$$

where C_0 is the capacitance without the dielectric, κ is the dielectric constant, L is the length, a is the inner radius, and b is the outer radius. The capacitance per unit length of the cable is

$$\frac{C}{L} = \frac{2\pi\kappa\epsilon_0}{\ln(b/a)} = \frac{2\pi(3.6)(8.85 \times 10^{-12} \text{ F/m})}{\ln[(0.40 \text{ mm})/(0.10 \text{ mm})]} = 1.44 \times 10^{-10} \text{ F/m} = 0.144 \text{ nF/m}.$$

39. (a) Before the switches are closed, the charge on the capacitors are:

$$q_1 = C_1 V = (1.0 \times 10^{-6} \text{ F})(200 \text{ V}) = 2.0 \times 10^{-4} \text{ C}$$

$$q_2 = C_2 V = (3.0 \times 10^{-6} \text{ F})(200 \text{ V}) = 6.0 \times 10^{-4} \text{ C}$$

After the switches are closed and the system gets a new equilibrium state, **the total charge on the bottom surfaces of the capacitors is conserved** and the potential difference from a to b is exactly the potential difference across C_1 as well as the potential difference across C_2 , so we have:

$$\left. \begin{aligned} q'_1 + q'_2 &= q_2 - q_1 = 4.0 \times 10^{-4} \text{ C} \\ \frac{q'_2}{C_2} &= \frac{q'_1}{C_1} \Rightarrow q'_2 = 3q'_1 \end{aligned} \right\}$$

$$\Rightarrow q'_2 = \frac{3}{4} \times 4 \times 10^{-4} \text{ C} = 3.0 \times 10^{-4} \text{ C}$$

$$q'_1 = \frac{1}{4} \times 4 \times 10^{-4} \text{ C} = 1.0 \times 10^{-4} \text{ C}$$

$$V_{ab} = \frac{q'_1}{C_1} = \frac{1.0 \times 10^{-4} \text{ C}}{1.0 \times 10^{-6} \text{ F}} = 100 \text{ V}$$

(b) The charge on capacitor 1 is now $1.0 \times 10^{-4} \text{ C}$.

(c) The charge on capacitor 2 is now $3.0 \times 10^{-4} \text{ C}$.

49. (a) According to Eq. 25-17 the capacitance of an air-filled spherical capacitor is given by

$$C_0 = 4\pi\epsilon_0 \left(\frac{ab}{b-a} \right). \quad \text{0.5 point}$$

When the dielectric is inserted between the plates the capacitance is greater by a factor of the dielectric constant κ . Consequently, the new capacitance is

$$C = 4\pi\kappa\epsilon_0 \left(\frac{ab}{b-a} \right) = \frac{6.91}{8.99 \times 10^9 \text{ N} \cdot \text{m}^2/\text{C}^2} \cdot \frac{(0.0120 \text{ m})(0.0170 \text{ m})}{0.0170 \text{ m} - 0.0120 \text{ m}} = 0.0314 \text{ nF}.$$

0.5 point

(b) The charge on the positive plate is $q = CV = (0.0314 \text{ nF})(73.0 \text{ V}) = 2.29 \text{ nC}$.

0.5 point

(c) Let the charge on the inner conductor be $-q$. Immediately adjacent to it is the induced charge q' . Since the electric field is less by a factor $1/\kappa$ than the field when no dielectric is present, then $-q + q' = -q/\kappa$. Thus,

$$q' = \frac{\kappa - 1}{\kappa} q = \left(\frac{6.91 - 1.00}{6.91} \right) (2.29 \text{ nC}) = 1.96 \text{ nC}.$$

0.5 point

52. We do not employ energy conservation since, in reaching equilibrium; some energy is dissipated either as heat or radio waves. Charge is conserved; therefore, if Switch S is first thrown to the left side until capacitor 1 reaches equilibrium, the charge on C_1 is Q , and q_1 , q_2 and q_3 are the charges on C_1 , C_2 and C_3 after the switch is thrown to the right and equilibrium is reached, then

$$Q = q_1 + q_2 + q_3. \quad \text{0.5 point}$$

Where $Q = C_1 V_{\text{bat}} = 120 \mu\text{C}$

0.5 point

Since the parallel pair C_2 and C_3 are identical, it is clear that $q_2 = q_3$. They are in parallel with C_1 so that $V_1 = V_3$, or

$$\frac{q_1}{C_1} = \frac{q_2}{C_2} = \frac{q_3}{C_3} \quad 0.5 \text{ point}$$

which leads to $2q_1 = q_3 = q_2$. Therefore,

$$Q = q_1 + 2q_1 + 2q_1 = 5 q_1$$

which yields $q_1 = Q/5 = 24 \mu\text{C}$.

0.5 point